Web Applications and Security

The big idea

This unit covers three things every computer user should know: **accessibility options** that make computers usable by everyone, **networking and the Internet** that connect us, and **online safety and security** that keep us protected.

Accessibility options

Accessibility means designing computers so people with disabilities (vision, hearing, movement, learning) can use them. In Windows: **Control Panel > Ease of Access > Ease of Access Center** (newer Windows: **Settings > Accessibility**).

Need	Option	What it does
Keyboard	**StickyKeys**	Press Shift, Ctrl, Alt one at a time instead of together
Keyboard	**FilterKeys**	Ignores brief or repeated keystrokes
Keyboard	**ToggleKeys**	Plays a sound when Caps Lock, Num Lock or Scroll Lock is pressed
Sound	**SoundSentry**	Shows a visual warning (flash) in place of sounds
Sound	**ShowSounds**	Displays captions for speech and sounds
Display	**High Contrast**	Uses strong contrasting colours for readability
Display	**Cursor options**	Changes cursor blink rate and width

Need	Option	What it does
Mouse	**MouseKeys**	Moves the pointer using the numeric keypad
Others	**Magnifier, Narrator, On-Screen Keyboard**	Zoom screen, read text aloud, type with a pointer

Serial Keys let alternative input devices be attached through a serial port.

Networking basics

A **network** is two or more computers connected to share data and resources (printers, Internet).

- **Peer-to-peer (P2P)**: all computers are equal; each can share. Good for small setups.
- **Client-server**: a central **server** provides services to **clients**. Secure and manageable for large setups.

Types by area:

- **PAN** – personal (Bluetooth between phone and earbuds).
- **LAN** – a room, building or campus.
- **MAN** – a city (cable TV network).
- **WAN** – countries/continents; the **Internet** is the largest WAN.

The **Internet** is a global network of networks. The **World Wide Web (WWW)** is a system of linked web pages accessed through the Internet using a browser. An **intranet** is a private network for an organisation; an **extranet** allows limited outside access.

Internet access options: dial-up (slow, uses phone line), **DSL**, **cable**, **satellite**, **3G/4G/5G mobile**, **Wi-Fi** (wireless LAN through a router/hotspot), **WiMAX**. Data travels in small **packets**, each carrying source and destination **IP addresses**.

Using web applications

- **Instant messaging** – real-time text chat (e.g. WhatsApp, Google Chat).
- **Blogs** – online journals with posts in reverse date order (Blogger, WordPress).
- **Online transactions** – shopping, banking, UPI payments; check for **https** and a padlock.
- **E-mail, cloud storage, online office tools**.

Online safety and security

Threats: **viruses, worms, trojans, spyware, phishing, identity theft, cyberbullying**.

SAFE HABITS

- Use **strong passwords**: at least 8 characters, mix of upper and lower case letters, numbers and symbols; do not use names or birthdays; do not share.
- Use **antivirus** and keep it updated; turn on the **firewall** (blocks unauthorised network access).
- Do not click unknown links or download from untrusted sites.
- **Clear browsing history and cookies**, especially on shared computers; use private browsing.
- Log out after use; avoid saving passwords on public computers.
- Share minimum personal information on social media.

Cookies are small text files a website stores on your computer to remember you; they can be misused to track you.

STRONG DOES NOT MEAN LONG ONLY

"aaaaaaaaaa" is long but weak. A strong password mixes character types, e.g. `Mn#47kuR!q`.

Occupational health and safety

Workplaces must reduce hazards: **ergonomics** (good chair, screen at eye level, regular breaks), **electrical safety**, **fire safety** (know exits; use the correct fire extinguisher), and **first aid**. Basic emergency steps: stay calm, raise the alarm, evacuate by stairs (not lifts), and report the accident.

COMMON 1-MARK QUESTIONS

"Which feature plays a sound when Caps Lock is pressed?" → ToggleKeys.
"Which shows visual warnings for sounds?" → SoundSentry.

Quick recap

- Accessibility: StickyKeys, FilterKeys, ToggleKeys, SoundSentry, ShowSounds, High Contrast, MouseKeys.
- Networks: P2P vs client-server; PAN, LAN, MAN, WAN; Internet ≠ WWW.
- Web apps: IM, blogs, online transactions.
- Safety: strong passwords, antivirus, firewall, clear cookies, think before clicking.

Practice questions

1. Which accessibility feature lets you press Ctrl, Shift and Alt one at a time?
   (a) StickyKeys (b) FilterKeys (c) ToggleKeys (d) MouseKeys
2. Which feature plays a tone when Caps Lock is pressed?
   (a) SoundSentry (b) StickyKeys (c) ToggleKeys (d) ShowSounds
3. In which network does a central computer provide services to other computers?
   (a) Peer-to-peer (b) Client-server (c) Personal area network (d) Bluetooth pairing
4. The largest WAN in the world is
   (a) an intranet (b) a LAN (c) a MAN (d) the Internet
5. Small text files stored by websites on your computer to remember you are called
   (a) packets (b) blogs (c) cookies (d) macros
6. Which of these is the strongest password?
   (a) password123 (b) 12345678 (c) riya2009 (d) Gt7#qL9!wz
7. SoundSentry is useful for people with
   (a) weak vision (b) hearing difficulty (c) difficulty using a mouse (d) difficulty pressing two keys together
8. Differentiate between a LAN and a WAN.
9. Write any six guidelines to stay safe online.

Answers: 1. a 2. c 3. b 4. d 5. c 6. d 7. b 8. see notes 9. see notes